

Short Form Truth Tables and Logical Equivalences

1 Short Form Truth Tables

Last time we saw how to construct a long form truth table.

Example. Construct a truth table for the formula $\sim P \& (P \supset Q)$.

P	Q	$\sim P$	$P \supset Q$	$\sim P \& (P \supset Q)$
T	T	F	T	F
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

1.1 Shortform truth tables

We create **short form truth tables** this:

1. Fill in all possible combinations of truth values for the atomic sentences which occur in your sentence, to the left of the sentence.
 - convention: list atomic sentences in alphabetical order
2. Copy over truth values for all atomic sentences in the sentence being diagrammed, (in a column under each occurrence of each atomic sentence).
3. Fill in the truth value for larger subsentences formed by applying a given logical operator (in a column under this operator), when you can (i.e., when you have truth values for their component sentence(s)).
4. Keep going until you reach the main logical operator, and have a column containing truth values for the whole sentence (under each valuation) has been produced. Circle this column.

The truth value you write under the **main connective** for your sentence gives the truth value for the **whole sentence** under that valuation. So we will circle that column/print it in red.

Demo: Make a truth table for $I \& (P \vee I)$

First we fill in all possible combinations of truth values for the atomic sentences.

I	P	$I \& (P \vee I)$	
T	T		
T	F		
F	T		
F	F		

Now we copy those truth values to the corresponding columns on the right hand side.

I	P	I	&	(	P	∨	I	)
T	T	T			T		T	
T	F	T			F		T	
F	T	F			T		F	
F	F	F			F		F	

Now we fill in the columns under the logical operators as soon as all the columns they apply to are filled in. So we start with $\vee$

I	P	I	&	(	P	∨	I	)
T	T	T			T	T	T	
T	F	T			F	T	T	
F	T	F			T	T	F	
F	F	F			F	F	F	

Now we can fill in the column for $\&$. This is the main connective, so you would circle it in your work.

I	P	I	&	(	P	∨	I	)
T	T	T	T		T	T	T	
T	F	T	F		F	T	T	
F	T	F	F		T	T	F	
F	F	F	F		F	F	F	

Q: Using the truth table above, what's an example of a valuation that makes this sentence true?
I: True, P : True or I: True P: False

2 Logical Equivalence

Two sentences $\mathcal{A}$ and $\mathcal{B}$ are **logically equivalent** iff, for every valuation, their truth values agree, i.e., if there is no valuation in which they have opposite truth values.

So we can test whether two sentences $\mathcal{A}$ and $\mathcal{B}$ are logically by setting up shortform truth tables for $\mathcal{A}$ and $\mathcal{B}$ side by side and then checking whether the columns holding the truth values for each sentence overall agree on each row.

Example. Show that $P \supset Q$ and $\sim P \vee Q$ are logically equivalent.

P	Q	P	$\supset$	Q	$\sim P$	$\vee$	Q
T	T	T	T	T	F	T	T
T	F	T	F	F	F	F	F
F	T	F	T	T	T	T	T
F	F	F	T	F	T	T	F

Since the columns for $P \supset Q$ and $\sim P \vee Q$ are identical, the two statements are logically equivalent.

3 Entailment

Remember the notion of informal validity from day one?

- Definition: An argument is **valid** iff it's impossible for all its premises to be true, while the conclusion is false.